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iv. Wind velocity: In still air, the surface above the stomata get saturated with water vapours and there is no need for more water vapour to come out. If the wind is breezy, water vapour gets carried away near leaf surface and DPD is created to draw more vapour from the leaf cells enhancing transpiration. However, high wind velocity creates an extreme increase in water loss and leads to a reduced rate of transpiration and stomata remain closed.

Activity

What will happen if an indoor plant is placed under fan and AC?

v. Atmospheric pressure: In low atmospheric pressure, the rate of transpiration increases. Hills favour high transpiration rate due to low atmospheric pressure. However, it is neutralized by low temperature prevailing in the hills.

vi. Water: Adequate amount of water in the soil is a pre-requisite for optimum plant growth. Excessive loss of water through transpiration leads to wilting. In general, there are three types of wilting as follows,

a. Incipient wilting: Water content of plant cell decreases but the symptoms are not visible.

b. Temporary wilting: On hot summer days, the freshness of herbaceous plants reduces turgor pressure at the day time and regains it at night.

c. Permanent wilting: The absorption of water virtually ceases because the plant cell does not get water from any source and the plant cell passes into a state of permanent wilting.

2. Internal factors

i. Leaf area: If the leaf area is more, transpiration is faster and so xerophytes reduce their leaf size.

ii. Leaf structure: Some anatomical features of leaves like sunken stomata, the presence of hairs, cuticle, the presence of hydrophilic substances like gum, mucilage help to reduce the rate of transpiration. In xerophytes the structural modifications are remarkable. To avoid transpiration, as in ***Opuntia*** the stem is flattened to look like leaves called **Phylloclade**. **Cladode** or **cladophyll** in ***Asparagus*** is a modified stem capable of limited growth looking like leaves. In some plants, the petioles are flattened and widened, to become **phyllodes** example ***Acacia melanoxylon.***

11.6.5 Plant Antitranspirants

The term **antitranspirant** is used to designate any material applied to plants for the purpose of retarding transpiration. An ideal antitranspirant checks the transpiration process without disturbing the process of gaseous exchange. Plant antitranspirants are two types:

1. To act as a physical barrier above the stomata

Colourless plastics, Silicone oil and **low viscosity waxes** are sprayed on leaves forming a thin film to act as a physical barrier (for transpiration) for water but permeable to CO_2 and O_2. The success rate of a physical barrier is limited.

2. Induction of Stomata closure

Carbon-di-oxide induces stomatal closure and acts as a natural antitranspirant. Further, the advantage of using CO_2 as an antitranspirant is its inhibition of photorespiration. **Phenyl Mercuric Acetate** (PMA), when applied